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A review of tracking concept drift detection in machine learning

Nail Adeeb Ali Abdu
Information Technology Department, Faculty of Engineering & Computing, University of Science and Technology, Aden, Yemen

Khaled Omer Basulaim
Information Technology Engineering Department, Faculty of Engineering, University of Aden, Jemen
ORCID ID: 0000-0002-4519-1133

ABSTRACT: The availability of time series streaming data has increased dramatically in recent years. Since the last decade, there has been a growing interest in learning from real-time data. While extracting significant information from data streams, online learning faces changes in data distribution. Concept drift refers to hidden data contexts that learning systems are unaware of it. Using previous training instances from the data stream, the classifier categorizes new instances. Hence in this regard, it is obvious to have considerable methods to discover the similarity or diversity status of the present streaming data that compared to the buffered data. This issue is defined and explored in contemporary literature as concept drift. This paper was initiated to identify the constraints of the contemporary methods devised to handle the concept drift in data streams in relation to supervised learning, we also review and categories the concept drift detectors with their key points. Eventually, the article presented observations can provide decision boundary detection at improved levels and discussing open research challenges and possible new research direction.

Keywords: concept drift, concept drift defensing, detection methods

1 INTRODUCTION

Data mining approaches are being used across multiple dynamic and real-time applications such as financial transactions, sensor networks, mobile communications, etc. However, the scalability of implementing these models in real-time applications which produce huge voluminous data in short periods of time is repeatedly challenged. As the information generated in such applications is dynamic and forms data-streams with voluminous instances, storing information prior to processing is not feasible. Further, specific applications require instant and real-time solutions. One of the most prominent issues faced by researchers during the mining process is that the data-streams in such instances can be non-stationary and witness concept drifts along the infinite length [1,2]. Due to the infinite distance and voluminous characteristics of these data-streams, storing information and utilizing the historical patterns for model training is not possible in the practical sense. Accordingly, new mechanisms that can process the information through a single scan and employ limited resources such as storage space and time are required. The several features of classical data mining over data stream mining are displayed in Table 1. In a stream with significant learning, the order of observation determines how the data are organized. Therefore, it is necessary to identify any changes in data distribution. One of the biggest problems with data stream mining is this detection, as the quality of prediction is affected if the changes is not noticed. [3]

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Table 1. Classical data mining vs data stream mining.

	Requirement of memory	Count of concepts	Count of passes	Requirement of time	Obtained of result
data stream mining	Limited	More than one	Single	Limited	Approximate
classical data mining	Unlimited	One	Multiple	Unlimited	Accurate

In addition, concept drifts are also observed is such real-time data streams where the underlying concept changes over time. The concept drift is observed whenever the underlying concept regarding the gathered information alters over some duration of stability. In the absence of the drift it is hard to maintain the balance between the most recent concept and the previous concept, which leads to form the stability-plasticity dilemma [4]. These shifts in the inflow data stream significantly impact the precision rates of the classification model because the classifier learns by historical stable data instances. Some of the areas where the concept drift phenomenon is observed include credit card fraud identification, spam recognition, and weather forecast. Accordingly, the concept drift identification process is considered as a vital phase [5]. The abstract view of model construction in the data stream is depicted in Figure 1. This figure consists of three models. The first module is the input module, which accepts instances or examples of training data as input from the algorithms. Because data streams instances are constantly arriving, the incoming data instances are stored in memory and processed later. The model predicts the labels of newly incoming data instances allowing us to maintain a high predictive performance over time. The aforementioned procedure will be repeated until the data are accessible. [6]

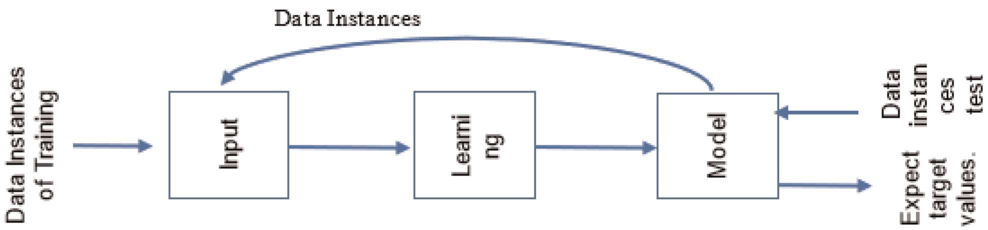

Figure 1. Construction a model in a data stream as an abstract view.

Upon detecting the concept drift in the data stream, the fixed classification models can yield inaccurate results [7]. For instance, consumer buying preferences can vary because of weather shifts or economic fluctuations. Accordingly, business intelligence entities and users of statistical prediction methods must identify such concept drifts within data-streams in a timely manner. The learning process for the classifier must be a continuous process as the model classifies the future instances by concepts learned. This continuous learning process ensures that any concept drift occurring in the streams can be tracked. The classifiers used for data-streams must be able to learn the patterns through a single scan and should be capable of performing classification at any time. Further, the model must be able to track the concept drifts over time and also preserve historical concepts. Typical classification of underlying concept drift should be by "speed of change". The change is considered as abrupt if the shift from one concept to the next concept occurs suddenly and in case the shift happens over few instances, it is considered as gradual. Alternatively, if the target concept is constant but data distribution shifts over time, it is considered as virtual [8]. However, in practical applications, both real and virtual concepts appear together [9]. The classification of drift as abrupt or gradual depends on its type upon observation. Sudden shifts over data-streams represent sudden-drifts, and on the other hand, slow transformation denoted gradual drift. An optimal classification model must be able to identify the concept drifts occurring in the data-streams to minimize accuracy loss identified during the concept-drift. Numerous methods are put forward to determine concept drift, and some of the prominent ones include online-algorithms, ensembles, and drift-detection models. Classification of error rates is the foundational concept for most of the drift detection methods [10]. Whenever any concept drift occurs then the trained classifier that belongs to the older data chunks becomes incompatible with the upcoming data chunks, and due to this the classified new chunk's error rate increases. Hence, error rate became the crucial parameter when detecting the concept drift occurrence.

2 CONCEPT DRIFT

The concept drift definition is flexible enough which can be defined informally depending on the use of the concept. For example, "the statistical properties of target concept changes over time in a situation" [11]. Finding the exact points of change can be very challenging, as the change between two distributions can be gradual and has to be differentiated from transient noise that can affect the stream. Consequently, change detection algorithms have to be robust (resistant to noise) and at the same time sensitive to true changes in the dataset, forming a classic stability-plasticity trade-off [12]. It is also essential for the algorithms to be able to reliably detect the drift with minimal delays, while also efficiently executing the detection in a shorter time than the next instance arrives and in a constant space complexity [13]. A learner should know the sequential positioning of examples in the stream [14], particularly in cases like the arrival of the samples with latency or in an out-of-order. The differentiation between the local and global drifts [15] is also identified based on the drift influence on the problem space or parts of the problem space. Over the years, the research works have proposed various methods and techniques to find concept drifts in the area of data streams [16]. The proposed techniques are classified into (1) windowing technique, (2) ensemble classifiers, and (3) drift detectors. Among these techniques, the windowing technique presents a manageable supporting mechanism for saving the latest data as well as updating the classifier of the system for maintaining accuracy. However, the size of the window hampers the drift finding process. Ensemble classifier gives the best chance to modify any of both methods, their classifier structure and aggregation methods based on the degradation of the main classifier performance because of concept drifts. A sliding window which is one of the most common windowing techniques saves the instances of the recent data in a limited number only. This Sliding window technique plus traditional learning algorithms can construct classifiers for data streams. The efficacy of drift detection is often downgraded, if fixed size sliding window technique is used. The small size window finds sudden changes quickly, however, drops stability accuracy (in periods) because the large size window gives away to see the swift changes occurring in the data streams [17]. In addition to drift detectors and windowing technique, and, for learning from data streams, the ensemble is also the most common technique. In ensembles, two types, Online ensemble, and Block-based ensemble are preferred to handle data changes. To update the classifier, online ensemble learns incrementally after every instance of data and a Block-based ensemble first processes the data which is blocked and then the classifiers are upgraded. Abrupt drifts are found by the online ensemble effectively, however, they are unable to detect the gradual drift, which appears during the data streamed over a considerable period of time. However, the Block-based ensemble finds the gradual drift efficiently but fails to detect sudden drift. The research work [17] presents Accuracy Updated Ensemble (AUE2). The concept drift of weighted base classifiers in the ensembles is adapted by the AUE2. Moreover, the AUE2 forecast class label, and records of small-sized block instances accurately. These records support for detecting the gradual drift demanding an exhaustive class labeling of these ensemble models. However, on the availability of the small amount labeled instances to build classifiers, the clustering technique is preferred. The work [18] presents ensemble classifiers as well as clusters for categorizing the test data with a perfect load as they communicate the class labels along with the class information on training data. In the work [19], using kernel density estimation for detecting the Twitter hotspot cluster activities as well as finding the hidden patterns and drifts in spatial trajectories that are chosen from spatio-temporal where the data is embedded with texts of twitter, Senaratne *et al.*, presented a framework. The majority of drift detection algorithms, which were proposed, failed in maintaining the high level of generalization between the two concepts, i.e., old and new ideas. Hence to address the issue, in the work [20] presented a system to save various diversity ensembles of the different levels. The high diversity ensembles and low diversity are joined together whenever the fresh ideas of the concepts are generalized with the existing plans of the idea. While dealing with the data streams, the detection of novelty is one of the challenges. In case of introducing either a new class or new feature to an old set of data streams, innovation is found. The general framework to handle the concept drift is explained in the Figure 2 dealing with the detection of knowledge from data collected from distributed systems and online data, which operates in a complex, dynamic and unstable environment. It explains the types of learners, their strategies and performance parameters.

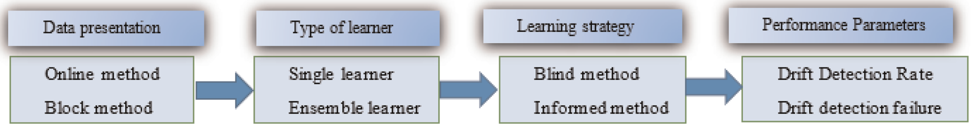

Figure 2. General framework to handle the concept drifts.

3 ENSEMBLE METHODS

The popular methods Bagging and Boosting, which are presented in the work [21] is used to improve the weak learners' algorithms accuracy. However, using different strategies, these methods utilize trained set classifiers on the training data and aggregate every classifier response; then it combines in an ensemble to give better predictions. Mainly, for training, bagging method uses various randomly produced bootstrap samples, through the repeated sampling of the training set. On the contrary, boosting method trained multiple classifiers, utilizing multiple training data distributions, provided the diversity of the classifiers and varies according to the earlier forecasting. However, attention is needed that several boosting algorithms give theoretical guarantees for their results. Likewise, the method Edwin-OZABOOST that included in MOA framework is preferred by ADWIN to detect the concept drift. In the work [22], the Leveraging Bagging named as LevBag is a newly featured version of Oza and Russell's Online Bagging. As a hard-coded concept drift detector, this Levbag is joined with ADWIN. Besides, this method presents two changes where the first change is to raise the diversity ($\lambda = 6$) value in the Poisson distribution then drives to a growth in the experts trained probabilities on the same instance. To change the instances prediction process used by the experts predict to rise in diversity and decrease the correlation in the second change. Accordingly, the LevBag with all these modifications provides better accuracies when compared to Edwin OzaBag method. Another method DDD presented in the works [20]. This is a default in the EDDM method. The researcher proclaims that the e-Detector makes better rates of the recall and false negative with minimal rise in the frequency of the false positive detection. Furthermore, it possesses a robust generalization capacity when compared with the detectors.

4 OBSERVATIONS

It is observed that many of the drift detection methods presume the complete accessibility of labeled streams to analyze data distribution as well as classifier performance. The work [23] offers an entirely unsupervised and semi-supervised approach to develop the applicability of the existing drift detection techniques supposing that streams are hardly labeled. The two methods emphasize on detecting imbalanced class distributions and statistical changes with the support of an adapted Page-Hinkley test. The work [24] depicts the dependency of their concept drift detector on data distribution change detector, instead of actual class labels. The contemporary studies emphasize the development of drift detection methods and techniques to address the real drift. Ample research hasn't been done in this area impacting the classification requirement as well as the performance. Notwithstanding the impact of drifts above on true decision boundaries, the research can provide decision boundary detection at improved levels.

5 CONCLUSION

The aim of the paper is to classify and confirm contemporary literature on data stream mining under the influence of conceptual drift. The review notes are that the current field of research contributes seriously to innovative and optimal drift detection methods for mining data streams. The review explored the existing report contributions to the literature of the last decade. The majority of these models target factors such as high-speed data flows, noisy data flows, mining accuracy, and time complexity of the mining strategy. Factors such as the context of the drift concept, the drift concept, the time validity, and the drift concept were ignored with recursive concepts. Going forward, it is clear to conclude that there is a lot of room for research in order to find new models to deal with understandable drift in mining data streams in the context of factors such as the concept of time drift, validity, context, and the impact of the iterative concept. Our contributions to future re-research will be the new or expanded models in the review competition.

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